

Validation-first AI

Deterministic vs Generative AI. Practical Skills for Research Professionals

June, 2026

Dr. Cody Dodd

LSE | Service Canada | ResearchAI

- 1) Intros and automation
- 2) Thesis – Deterministic versus generative logic
- 3) AI Ensembles
- 4) Showcase 1 – GenAI and Math
- 5) Showcase 2 - GenAI and Tagging
- 6) AI Enablement

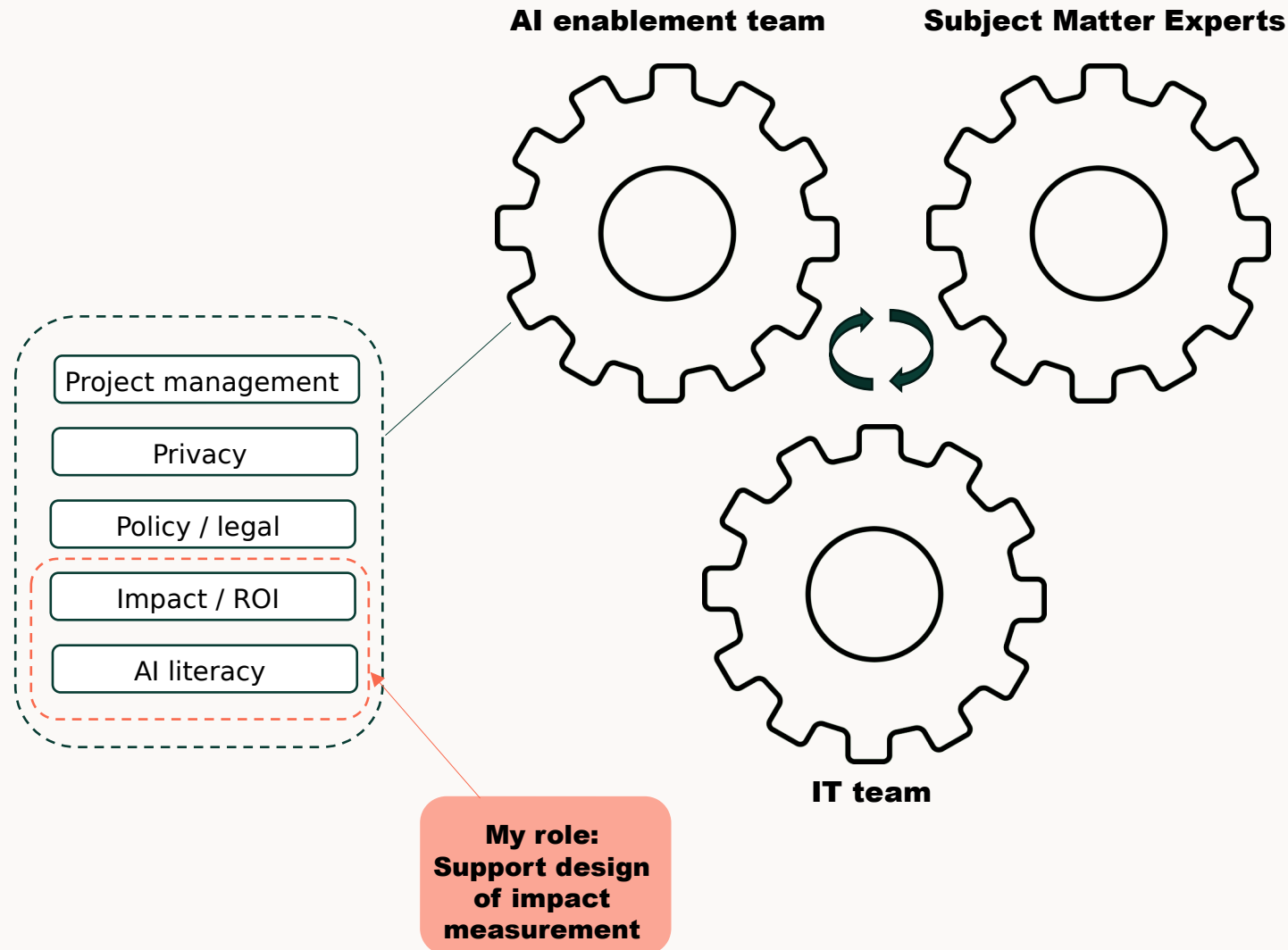
“Today is about learning how to use AI confidently... by grounding generative systems in deterministic logic.”

Introducing Cody Dodd - Relevant work

London School of Economics (LSE): Ongoing research and collaboration on negotiated governance - how different algorithmic artefacts reshape control, accountability, and institutional design.

- Management Department
 - School of Public Policy
 - Methodology Institute
 - Digital Skills Lab
 - Data Science Institute
-
- **Public sector practice:** Directly support the governments and NGOs across Canada and the UK in deploying AI and automation through governance-first, privacy-aware architectures.
 - Institute for Citizen Centred Services
 - Federal (Service Canada), provincial/regional, municipal governments in Canada and UK
 - Canadian Research Insights Council

Introducing Dr. Cody Dodd - My typical role



PART ONE: Intros and Warm-Up (10 min) — “Where is AI already showing up?”

- From literature, architecture and anecdotes

PART TWO: Deterministic versus generative logic

- Where do you see AI in your organization today?
- Where do you wish it could help?"

The risks we know

Poor AI governance = Democratic failure

- **95% of AI investments fail to deliver** ([MIT Labs](#) + [HBR](#))
- **41% of workers encounter workslop monthly** (fake or empty outputs), trust lost for public institutions.
- **Only 3 out of 45 real sources** in a major study - [KPMG](#)
 - https://app.gptzero.me/documents/45f1e69a-991c-44d8-ad05-48036fba5085?scanType=bibliography_scan

Non-deterministic.

- Anyone who tells you a GPT can be trained to avoid mistakes, or that it just needs ‘better prompting’, is misunderstanding their architecture.
 - “It ignored one of my requests no matter how many times I asked it not to”
 - “It erased data it shouldn’t have”
 - “It summarized only parts”
 - “It made that one up”

“Context is a Goldilocks zone. Too little, it guesses, too much, it gets confused or takes shortcuts”

GPT benefits

Reformatting

Combine, summarize, tighten, change formats, add documentation.

Faster starts

Remove that slow start / blank page.
Create quick drafts.

Makes us more creative.

'Have you thought of?'

Quick pass to find spelling errors, pointers, change language style.

Faster coding

Generate scripts, build quick pilots, describe other code.

GPT challenges

- Difficult to capture ROI of GenAI as a productivity tool.
- Weak signals get amplified (with bias implications)
- Ungoverned systems hide accountability
- Algorithmic pipelines decide who and what gets heard
- Retrieval tends to skip core sections
- Not to mention data sovereignty, prohibitive costs, environmental impact

Regulatory tension

These regulatory contexts:

- GDPR; EU AI Act; Canada's AIDA (Bill C-27) and "Guiding Principles for the Use of AI in Government"; UK's ICO guidance and core AI principles
- expect AI to be:
 - 1) **Explainable** (You can show why an output happened)
 - 2) **Traceable** (You can show what the system did, step-by-step)
 - 3) **Human-oversighted** (A person can review, correct, and override)
 - 4) **Purpose-limited & data-minimal** (Only the necessary data, used for a clear purpose)

Regulations assume systems behave predictably — GenAI doesn't.

Comparison

	LLMs	Deterministic ML
Determinism	Non-deterministic — same prompt can yield different outputs	Predictable — same inputs always produce the same outputs
Grounding	No grounding — outputs are generated from patterns in text, not computation	Grounded — outputs come from explicit mathematical models
Reproducibility	Low — temperature, sampling, context drift, and hidden heuristics change results	High — fixed algorithms, fixed seeds, fixed data
Auditability	Weak — internal reasoning is opaque and cannot be inspected	Strong — every step is inspectable, traceable, and explainable
Error Behavior	Silent failure — confident but wrong answers	Transparent failure — errors are measurable and debuggable
Data Use	Trained on broad text; not tied to your dataset unless explicitly given	Trained only on your dataset; behavior is fully controlled

Why LLMs Behave This Way

Why LLMs Behave This Way (Architecture Explanation)

LLMs are next-token predictors, not reasoning engines. Their architecture creates these limitations:

- **They generate text, not truth.**
Every output is a probability distribution over words, not a computation.
- **They have no internal concept of “correctness.”**
They optimize for plausibility, not accuracy.
- **They do not execute algorithms unless explicitly connected to tools.**
Without Python or deterministic components, they simulate math instead of doing math.
- **Their outputs depend on hidden sampling parameters.**
Temperature, top-p, and context windows introduce randomness.
- **They cannot trace or justify their internal steps.**
The transformer architecture does not expose intermediate reasoning.

This is why LLMs are powerful narrators, but unreliable predictors.

Deterministic ML models:

- Compute, they don't guess
- Use fixed mathematical functions
- Produce the same result every time (if configured to)
- Are fully inspectable (coefficients, feature importances, SHAP, etc.)
- Can be version-locked, audited, and certified
- Are required for regulated environments (finance, government, health)

PART THREE: Ensemble AI - Combining deterministic and non-deterministic logic

- Text analysis example
- Triaging communications, requests, and taskings
- Other examples

Validation-first AI



Start with deterministic systems:

- Classic tools like z-scores, crosstabs, regression still play a role. These can predictably cluster data, detect outliers, and build grounded truth using rules-based logic.
- For text analysis: “Voice of Citizen”:
 - Keyword-based matching – Most trusted, least flexible
 - Include lemmas and other language rules – Still predictable, more flexible
 - Matches based on token similarity – Less predictable, more flexible
 - Machine-learning based on human tagging – More predictable, more flexible, but requires training.

Once truths are established, turn generative AI and agents:

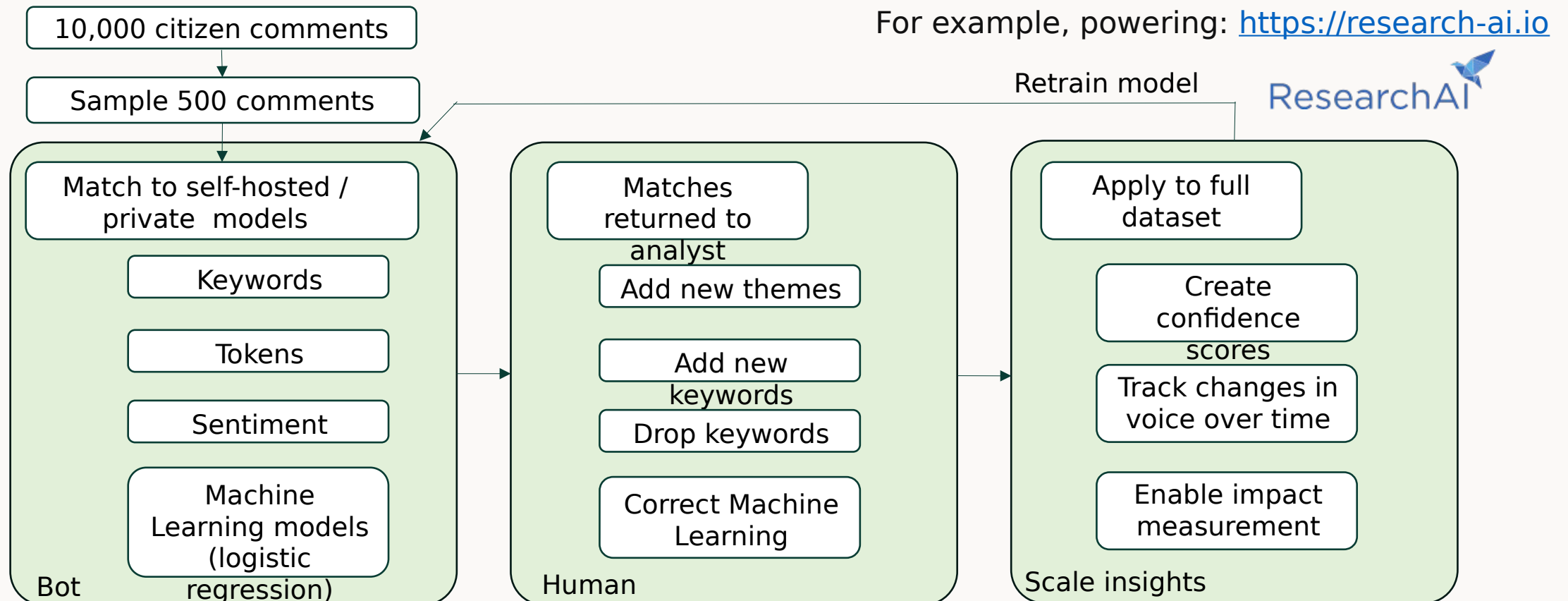
- Summarization on top of grounded truth.
- Treat as suggestive, not authoritative.

Ensure steps produce metadata:

- Track what algorithms did what when.

Methodology: Validation-first text analysis AI

Canadian Federal department, Yukon Government, Peel region, and dozens of other governments: capturing voice and building trust using governed architecture. **Thousands of citizen comments analyzed a month, with transparent, reproducible tags of exactly what is being said.** This is an example of [validation-first AI](https://research-ai.io).



Methodology: Validation-first text analysis AI

Example of Validation-first-AI supporting theme tracked over time



Saved themes: 17



[Edit themes](#)

[Generate themes](#)



Staff - Compliment

49.0%
(400 / 816)



24 keywords

helpful staff friendly knowledgeable personnel officer pr

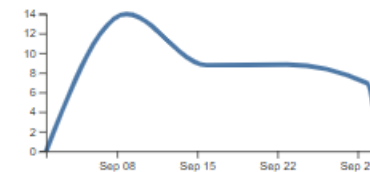
1 notes

Great office. Thanks



Overall satisfaction

9.4%
(77 / 816)



13 keywords

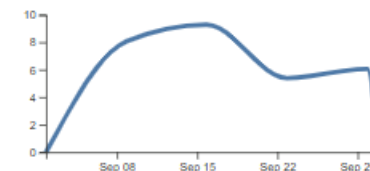
great experience excellent service awesome experience tout était par

0 notes



Ease / Process - Compliment

7.1%
(58 / 816)



4 keywords

painless efficient smooth easy

0 notes

Methodology: Validation-first text analysis AI

Who is using this:

- Yukon Government now tracks emergency services in real-time, redistributing resources based on citizen voice.
- Canada's largest service department validates web and operational data using voice of client.
- Parliamentary committees are tracking changes in parliamentary priorities from transcripts.

Examples of how to drive citizen engagement and trust:

- When a new program is launched, set target for change in themes
 - “A consistent top complaint is difficult authentication for the digital service (10% of comments each month), the new log-in feature should decrease this to 5%.”
- Automatic red-flag reports when certain topics reach certain thresholds
 - “Complaints about fraud should never exceed 2% of all comments”

Methodology: Validation-first text analysis AI



Ingestion and cleaning (deterministic)

Similarity and clustering (deterministic)

Human validation (governance)

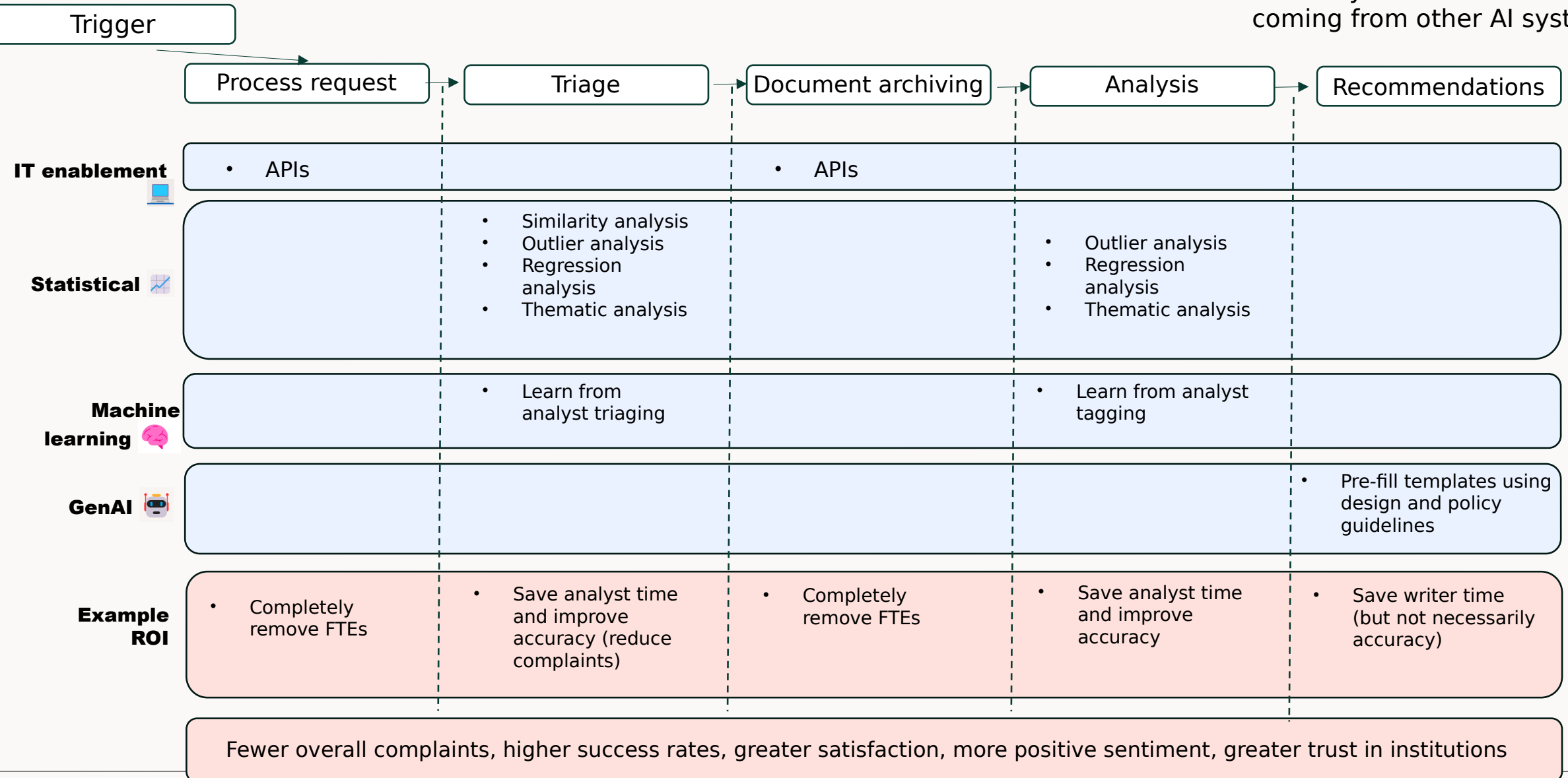
Machine learning from human actions (deterministic, negotiated logic)

GenAI summarization and narrative building (constrained)

Transparent reporting (reliable insights retraceable back to citizen comments)

Example: Triage automation solutions

Notice: very little GenAI: most value coming from other AI systems



AI enablement elements

	What It Does	Strengths	Weaknesses	When to Use
IT Enablement	APIs, integrations, workflow triggers, routing, data access	Reliable, scalable, zero hallucinations	Requires engineering	Always first — foundation for automation
Statistical / Deterministic	Rules, thresholds, similarity, z-scores, crosstabs	Predictable, auditable, explainable	Less flexible	Intake, triage, validation, early classification
Machine Learning	Logistic regression, Random Forests, clustering	Accurate, reproducible, learns from humans	Needs training data	Tagging, prioritization, risk scoring
Generative AI	Drafting, summarizing, narrative building	Fast, expressive, creative	Non-deterministic, not auditable	Often last — after truth is established

Output: Automation use-cases

	AI Service Capability (Primary Problem)	What this is	AI solution
1	Request Intake & Triage (single door)	(Channel-agnostic but not channel optional) Multi-channel, compliant request intake with early completeness validation, request classification (simple/complex), risk-based triage, categorisation, prioritisation, and controlled nudging toward preferred digital channels.	Deterministic workflows to detect likely categories for triaging and auto-fill forms. No GenAI or agents needed
2	Tasking, Tracking & Routing (orchestration)	End-to-end orchestration of work items across systems, including assignment, status visibility, escalation, exception handling, and audit-ready traceability.	Deterministic workflows for detecting likely tasks and routing. No GenAI needed, but agents could support.
3	Drafting & Content Creation	First-draft creation using authoritative content sources and templates, supported by pre-draft redaction where required, with mandatory human validation and version control	GenAI to consume templates to provide suggestive responses. Humans must finish the draft. Deterministic 'similarity' analysis to provide suggestions.
4	Governance & Committee Support	End-to-end governance enablement including intake, records, metadata, decision traceability, and scalable onboarding across formal and informal committees.	CMS and data access is the main challenge, not automation. The solution explored better utilizing existing CMS systems.
5	Transcription & Summarisation	Secure meeting capture, assisted summarisation, and records-of-decision creation, with human validation, language handling, and retention controls.	Summarization and transcription speeds up archiving but French vs. English translations still struggle.
6	Information Storage, Access & Retrieval	Discovery, retrieval, preparation, and secure delivery of authoritative information and records across repositories, including digitisation, deduplication, sensitivity handling, and controlled external access.	Data access remains a main challenge (similar to area 4). API development can produce more savings.

PART FOUR: Showcase and play

- Machine learning
- Drivers analysis
- Text analysis

PW: CRIC2026

Github: <https://github.com/codydodd/validation-first-ai-workshop>

Email me for slides: cody.d@researchai.io

Activity 1 — Predicting Diabetes (Deterministic ML)

This activity introduces deterministic ML using the Pima Indians Diabetes dataset.

What to observe:

- Deterministic ML produces stable, reproducible predictions.
- Compare Python results to an LLM asked to predict diabetes without labels.
- Then compare again when the LLM is given labeled data.
- Try the same with the larger bank dataset.

Notice that some LLMs (e.g., Claude) get close only because they run Python tools, not because they “understand” the data.

This demonstrates the difference between simulation and computation.

Claude is trained to turn to predictable, deterministic tools. It is using LLMs to 'orchestrate' but not 'compute'.

Binary classification predictions for CSV data ▾

experience with government online social security applications.

You will:

1. Read all responses I provide.
2. Identify a set of discrete, non-overlapping, service-experience themes.
3. Create a final theme list that:
 - Is concise (15–40 themes)
 - Uses short, clear labels

Show more

Jun 14 ↻ ✎ 📄

Verifying zero-theme rows and scrutinizing high-frequency theme precision >

Now adding the Complaint pattern, Application Status "issued or not", and verifying the duplicate `key for...log` line doesn't remain:

Edited 2 files, ran a command ▾

🔍 Check for duplicate key for log pattern

Script

Classify v3 · PY

Copy ▾ ↻ ×

```

1  import pandas as pd
2  import re
3  import unicodedata
4
5  df = pd.read_excel('/mnt/user-data/uploads/text_analysis_online_social_security_application_no-results.xlsx')
6
7  def normalize(text):
8      text = str(text)
9      text = text.replace("'", '"').replace('\"', '')
10     nfkd = unicodedata.normalize('NFKD', text)
11     no_accent = ''.join(c for c in nfkd if not unicodedata.combining(c))
12     t = no_accent.lower()
13     return t
14
15  def augment(t):
16     """Add canonical tokens for common spelled-out / misspelled benefit names
17     to make downstream pattern matching robust, without destroying original text."""
18     extra = []
19     gis_patterns = [
20         r'guarante?e?d?s+income\s+suppl?ement', r'guareanteed income supplement',

```

Activity 2 — Drivers Analysis with Random Forests

Start with the Wine Quality dataset, then the Airline Satisfaction dataset.

What to observe:

Random Forests provide ranked feature importances.

- Compare these to LLM-generated “drivers.”

Some LLMs get close only when they execute real Python, not when they guess. Others hallucinate drivers entirely.

Activity 3 — Predicting Themes (Dictionary vs ML vs LLM)

This activity uses a dataset of ~900 human-tagged comments.

What to observe:

- Run a dictionary-based topic model using custom_topic_model-online_application.xlsx.
- Then run a machine learning classifier trained on human tags.
- Compare both to a pre-tagged dataset from a more complex ML model.
- Finally, ask an LLM to replicate the tagging.

We see:

- Deterministic dictionary methods are transparent but brittle.
- Generic ML models are not necessarily as accurate as keywords, without extensive coding.
- More advanced / managed ML models can get remarkably accurate with limited tagging.
- Without python, LLMs are inconsistent, often drifting or hallucinating categories.

This demonstrates the limits of generative systems for structured classification.

The power of typical LLMs

Copilot followed our instruction:

“You are a binary classifier.”

But what it actually did was:

- read each row as text
- apply loose, heuristic pattern-matching
- guess based on correlations it has seen in training
- output 0 or 1 based on those heuristics

It did not:

- train a model
- optimize weights
- evaluate accuracy

It simply simulated the behaviour of a classifier.

This is why copilot can feel like it's doing ML — but it isn't.

Why the manual deterministic model may be better

- it actually trains
- it optimizes weights
- it learns nonlinear boundaries
- it is consistent
- it is reproducible

The power of Claude

	Typical LLMs	Claude
Tool Use	Rarely uses tools reliably	Strong, consistent tool-calling (Python, retrieval, code execution)
Numeric Reasoning	Simulates math → errors	Delegates math to Python → grounded results
Following Instructions	Drifts, shortcuts, ignores constraints	More disciplined, more literal
Hallucination Control	High	Lower (but not zero)
Reproducibility	Low	Higher, but still non-deterministic

Claude is still a generative model, but its architecture and training emphasize:

- Tool orchestration over simulation
- Delegation over guessing
- Grounding over plausibility
- Refusal over hallucination

This makes Claude behave closer to a validation-first system — especially when it chooses to run Python instead of predicting.

Even with strong tool-use:

- It can still ignore a tool call
- It can still hallucinate when uncertain
- It can still drift with long context
- It can still produce different outputs across runs
- It cannot be audited internally
- It cannot be version-locked by the user

Claude is the best of the generative models for structured tasks — but it is still a probabilistic narrator, not a deterministic engine.

Beyond LLMs – the power of Agents and Skills

Agents and skills introduce deterministic structure around an LLM. They reduce drift by giving the model tools and rules instead of letting it improvise

	LLM Alone	Agent / Skill Layer
Task Execution	Simulates steps	Executes real steps (Python, APIs, workflows)
Consistency	Variable	Deterministic nodes + repeatable logic
Memory of Process	Weak	Explicit plans, state, and checkpoints
Error Handling	Unpredictable	Rules, retries, fallbacks
Data Access	Limited to prompt	API calls, databases, validated inputs
Auditability	Opaque	Logs, traces, reproducible runs

Agents don't make LLMs deterministic — but they wrap them in deterministic components:

- Python nodes
- API calls
- Validation steps
- Rule-based checks
- Workflow engines

This creates a hybrid system: LLM for language → deterministic tools for everything else.

PART FIVE: AI enablement

- Identify a process (Intake? Triage? Coding? Drafting? Etc.)
- Break into steps
 - Is this data access? *IT Enablement*
 - Is this validation or classification? *Deterministic*
 - Is this pattern learning? *Machine learning*
 - Is this drafting or summarizing? *GenAI*

PART FIVE: Concluding thoughts

Take-aways for AI and democratic insights

AI works best as a system, not a tool:

- Governance privacy, policy, IT, and SME's must co-own it
- Be sure to have AI literate voices at the table (who can tell the difference between deterministic and generative systems, who understand sources of bias and hallucinations, etc)

Deterministic automation delivers the biggest ROI:

- Triage, routing, validation, and intake outperform GenAI for reliability, ease-of-adoption, and impact.

GenAI is powerful but fragile:

- Best used for drafting, summarizing, and templated content after validation.
- Can find compelling narratives, but not strong at maintaining good sources.

Validation-first can produce reliable measures of citizen voice:

- Self-hosted models that rely on deterministic logic (avoiding hallucinations and fabricated comments) help us trust text analysis, and in turn, drive policy and service improvement.

Governance-first teams outperform tech-first teams:

- because they understand risk, accountability, and citizen-impact

Transparency builds trust:

- When citizens see how their input was analyzed, trust increases measurably.

AI Automation enablement

Different algorithms do different things

- Be confident enough to ask when should we use a deterministic process versus a generative one
- **Risk can be managed**
 - We don't need a one size-fits-all solution. We can manage risks by building an ecosystem of different systems
- **Teams can be empowered**
 - Yes agents, but don't forget python.
 - Code-camps for non-coders + GenAI for code generation has led to teams finding programming superheroes they didn't know they had

Copilot alone isn't enough.

Advantages:

- Faster deployment of AI with real results
- ROI measured in down-stream savings and improved citizen trust and engagement

Take-aways for AI in better insights

AI that aligns with market research values looks like this:

- Predictable systems (deterministic steps you can explain and audit)
- Human-validated insights (voice interpreted with care, not hallucinated)
- Data-sovereign architectures (on-premise, transparent, privacy-first)
- GenAI used safely (as a narrator on top of validated truth, not a decision-maker), or if properly managed, as an orchestrator

Why it matters for democratic institutions:

- Clearer citizen voice → better policy signals
- Traceable pipelines → higher institutional trust
- Transparent methods → stronger legitimacy
- Reliable insights → foundations for deliberative democracy

Future work and contacts

From bureaucracy to a more deliberative ballot:

Advancing voice of client from service delivery, policy, and parliamentary management to supporting parliamentarians, NGOs, and other democratic and community institutions to incorporate citizen voice in reliable, transparent ways. This could open the door for deliberative democracy.

Governing generative platforms:

Upcoming research on how generative systems re-write the platforms literature which once assumed stable cores. Cores are now dynamic, and once-dynamic periphery features are now becoming stable to act as 'guard rails' for unpredictable cores.

Let's continue the conversation! If you're working on governing AI, citizen insight, or automation at scale—and want to avoid black-box decision-making—I'd welcome the conversation.

Dr. Cody Dodd

Researcher, instructor & practitioner in AI governance

🔗 LinkedIn: <https://codydodd.ca>

✉ Email: cody@codydodd.ca